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**THE SHAPE AND CONCENTRATION OF DARK MATTER HALOS
IN A Λ CDM COSMOLOGY**

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Abstract

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We use the Bolshoi, Bolshoi-Planck and Multidark cosmological simulations to study the statistics of dark matter halo concentrations and shapes in a Λ Cold Dark Matter (Λ CDM) cosmology. Dark matter halo concentration is compared using results from the Rockstar Phase-Space Temporal halo finder and the Bound Density Maximum halo finder with our analysis spanning the mass range $(6.3 \cdot 10^{10} - 3.2 \cdot 10^{15})h^{-1}M_{\odot}$ and $z=0-6$. We define concentration with two different scale radii by fitting a NFW (Navarro-Frenk-White 1997) profile and using a method presented in Prada et al. 2011, finding the average concentration lower by fitting NFW. Our results agree with previous results (Neto et al 2007, Prada et al. 2011, Ludlow et al. 2012) that halo concentration declines monotonically for $z=0-1$ and exhibits an upturn in the high mass range. For $z=2-6$ our results for average concentration start flat and display a rapid upturn in agreement with Prada et al. 2011. We impose relaxation criteria on our halos to reduce our sample to halos in dynamical equilibrium. We observe an increase in concentration from $z=0-2$ and an upturn for $z=0-1$, which is at odds with Neto et al. 2007 and Ludlow et al. 2012. We present results for the median halo shape using algorithms presented by Allgood et al. 2006 and Zemp et al. 2011 at R_{vir} and find the mean halo shape to decrease monotonically with increasing mass and redshift agreeing with Allgood et al. 2006. We find the Allgood et al. 2006 method to be more robust and not experience resolution effects from calculating shapes with a small number of particles.

The most elongated and spherical halos are found to be those with the lowest and highest concentrations respectively.

Contents

List of Figures	vii
List of Tables	ix
Dedication	x
Acknowledgements	xi
1 Introduction	1
1.1 A Brief Preamble	1
1.1.1 Cosmology	1
1.1.2 Dark Matter	3
1.1.3 Dark Matter Halos	6
1.1.4 Dark Matter Halo Shapes: Simulations	6
1.1.5 Dark Matter Halo Shapes: Observations	8
1.2 In This Thesis	8
2 Simulations	11
2.1 Numerical Simulations	11
2.2 Halo Finders	12
3 Dark Matter Halo Concentration	14
3.1 Halo Concentration: Rockstar	16
3.2 Halo Concentration: BDM	18
3.3 Dark Matter Halo Selection Criteria: Rockstar	19
3.3.1 Selection Statistics	23
3.4 Dark Matter Halo Selection Criteria: BDM	25
3.4.1 Selection Statistics	27
4 Dark Matter Halo Shapes	31
4.1 Rockstar Shape Calculation - Zemp et al. 2011	32
4.2 BDM Shape Calculation - Allgood et al. 2006	34
5 Summary and Discussion	38

Bibliography**41**

List of Figures

1.1	Dark matter halo of mass $3 \cdot 10^{13} h^{-1} M_{\odot}$ in the Bolshoi-Planck simulation. .	7
3.1	Halo mass-concentration relation for distinct halos for Bolshoi (open symbols) and Multidark (filled symbols) the NFW fitted scale radius.	15
3.2	Halo mass-concentration relation for distinct halos for Bolshoi (open symbols) and Multidark (filled symbols) using the Klypin scale radius.	16
3.3	Halo mass-concentration relation for distinct halos for Bolshoi (open symbols) and Bolshoi-Planck (filled symbols) using the Klypin scale radius and the BDM halo finder.	18
3.4	Halo mass-concentration comparison between all the halos and the relaxed sample. The relaxation criteria in this figure is $\kappa < 1.3$. Squares represent the relaxed samples and circles represent the whole simulation. Bolshoi is open symbols and Multidark is closed symbols.	19
3.5	Halo mass-concentration comparison between all the halos and the relaxed sample. The relaxation criteria in this figure is $x_{off} < 0.07$. Squares represent the relaxed samples and circles represent the whole simulation. Bolshoi is open symbols and Multidark is closed symbols.	20
3.6	Halo mass-concentration comparison between all the halos and the relaxed sample. Both relaxation criteria are used where $\kappa < 1.3$ and $x_{off} < 0.07$. Squares represent the relaxed samples and circles represent the whole simulation. Bolshoi is open symbols and Multidark is closed symbols.	21
3.7	Halo mass-concentration comparison between all the halos and the relaxed sample. The relaxation criteria in this figure is $\kappa < 1.1$. Squares represent the relaxed samples and circles represent the whole simulation. Bolshoi is open symbols and Multidark is closed symbols.	22
3.8	Halo mass-concentration comparison between all the halos and the relaxed sample. The relaxation criteria in this figure is $x_{off} < 0.035$. Squares represent the relaxed samples and circles represent the whole simulation. Bolshoi is open symbols and Multidark is closed symbols.	23
3.9	Halo mass-concentration comparison between all the halos and the relaxed sample. Both relaxation criteria are used where $\kappa < 1.1$ and $x_{off} < 0.035$. Squares represent the relaxed samples and circles represent the whole simulation. Bolshoi is close symbols and Multidark is closed symbols.	24

3.10	Percentage of relaxed halos ($x_{off} < 0.07$) for a given mass bin. Lines are drawn at $N_p = 5000$ for Bolshoi and Multidark. Bolshoi is open symbols and Multidark is closed symbols.	25
3.11	Percentage of relaxed halos ($x_{off} < 0.035$) for a given mass bin. Lines are drawn at $N_p = 5000$ for Bolshoi and Multidark. Bolshoi is open symbols and Multidark is closed symbols.	26
3.12	Percentage of relaxed halos ($x_{off} < 0.035$) for a given mass bin in the Bolshoi simulation. Squares represent the relaxed samples and circles represent the entire simulation.	27
3.13	Percentage of relaxed halos ($x_{off} < 0.035$) for a given mass bin in the Bolshoi-Planck simulation. Squares represent the relaxed samples and circles represent the entire simulation.	28
3.14	Percentage of relaxed halos ($\kappa' < 0.1$) for a given mass bin in the Bolshoi simulation. Squares represent the relaxed samples and circles represent the entire simulation.	28
3.15	Percentage of relaxed halos ($\kappa' < 0.1$) for a given mass bin in the Bolshoi-Planck simulation. Squares represent the relaxed samples and circles represent the entire simulation.	29
3.16	Percentage of relaxed halos ($x_{off} < 0.035$) for a given mass bin in the Bolshoi and Bolshoi-Planck simulations. Squares represent the Bolshoi simulation and circles represent the Bolshoi-Planck simulation.	30
3.17	Percentage of relaxed halos ($\kappa' < 0.1$) for a given mass bin in the Bolshoi and Bolshoi-Planck simulations. Squares represent the Bolshoi simulation and circles represent the Bolshoi-Planck simulation.	30
4.1	Median axis ratio at R_{vir} using the Zemp et al. 2011 method. We indicate where $N_p = 5000$ for Bolshoi and Multidark. Bolshoi is open symbols and Multidark is closed symbols.	32
4.2	Median axis ratio at R_{vir} using the Allgood et al. 2006 method. Bolshoi and Multidark. Bolshoi is open symbols and BolshoiPlanck is closed symbols.	34
4.3	Concentration-Mass relation for the 100 most elongated (circles) and 100 least elongated (crosses) halos at each redshift in the Bolshoi-Planck simulation.	35
4.4	Concentration-Mass relation for the 100 most elongated (circles) and 100 least elongated (crosses) halos at each redshift in the Bolshoi simulation.	36

List of Tables

2.1	Parameters for Λ CDM cosmological simulations. L_{box} is the length of one side of the simulation, N_p is the number of particles in the simulations, ϵ is the force resolution in a comoving (constant in time) coordinate system, M_p is the mass resolution (mass of each particle). The cosmological density parameters for the simulations are: Ω_m , Ω_b , Ω_Λ . The spectral index of the primordial power spectrum is n_s , h is the Hubble constant at the present time in units of 100 km/sMpc^{-1} and σ_8 is the amplitude of mass density fluctuations of spheres of $8 \text{ Mpc}h^{-1}$ at redshift $z=0$. Millennium parameters from Springel et al. 2005, Millennium-II parameters from Boyle-Kochlin et al. 2009, Bolshoi parameters from Klypin et al. 2009 and MultiDark parameters from Prada et al. 2011	12
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To The Universe,
thanks for being you.

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1

Introduction

1.1 A Brief Preamble

1.1.1 Cosmology

Physical cosmology seeks to understand the evolution, origin, and fate of the universe and explain the natural processes that arrange our universe the way it is today. The magnificence of such study and thought makes the field vibrant, interesting, and far-reaching. The birth of physical cosmology is often considered to be 1929, when Edwin Hubble discovered that the universe is expanding. This was found by observing that the recession velocities of galaxies followed the famous relation now known as Hubble's Law

$$V = H_0 d \tag{1.1}$$

where V is the recession velocity, d is the distance to the galaxy, and H_0 is the Hubble constant at the present time. From then on, cosmologists have been constructing models of the universe in accordance with physical observation.

To describe the evolution of our universe we must assume the Cosmological Princi-

ple stating the homogeneity and isotropy of the spatial part of spacetime. We then use the famous Einstein field equations describing how the mass-energy of matter curves spacetime

$$R^{\mu\nu} - \frac{1}{2}g^{\mu\nu}R + g^{\mu\nu}\Lambda = \frac{8\pi G}{c^4}T^{\mu\nu} \quad (1.2)$$

where $R^{\mu\nu}$ is the Ricci curvature tensor, R is the Ricci curvature scalar, $g^{\mu\nu}$ is the metric tensor, Λ is the cosmological constant, G is Newton's gravitational constant, c is the speed of light in a vacuum, and $T^{\mu\nu}$ is the stress-energy tensor. One must note that to be consistent with the homogeneity and isotropy of space, the universe can be modeled as a perfect fluid such that the stress-energy tensor for curved spacetime is

$$T^{\mu\nu} = (\rho + p)u^\mu u^\nu + g^{\mu\nu}p \quad (1.3)$$

where ρ is the density, p is the pressure, and u is the four-velocity of the fluid. Equipped with the proper metric tensor we can begin to explain the evolution of the universe.

This leads us to consider the Freidmann-Lemaitre-Robertson-Walker (FLRW) metric for expanding space as a solution to Einstein's equation

$$ds^2 = -dt^2 + a^2(t)\left(\frac{dr^2}{1 - kr^2} + r^2(d\theta^2 + \sin^2\theta d\phi^2)\right). \quad (1.4)$$

In the FLRW metric, k is the measure of the curvature of the universe which can take the value 0 for a flat geometry, +1 for an elliptical or closed geometry, and -1 for a hyperbolic or open geometry. We now know through observation that our universe today is spatially flat with small observational uncertainties, corresponding to $k = 0$. The coordinates (t, r, θ, ϕ) describe a spherically symmetric comoving spacetime. The parameter $a(t)$ is the cosmological scale factor which relates a fixed distance r in a comoving coordinate system (necessary for describing the expansion of the universe) to proper distance d by $d = a(t)r$. The scale

factor can be related to the cosmological redshift z :

$$a(t) = \frac{1}{1+z}. \quad (1.5)$$

By solving Einstein's equation for the FLRW metric we can then derive the exalted Freidmann Eq.(1.6, 1.7) describing the evolution of the scale factor through time:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho - \frac{kc^2}{a^2} \quad (1.6)$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left(\rho + \frac{3p}{c^2}\right). \quad (1.7)$$

Describing the evolution of the scale factor over cosmological time with the Freidmann equations allows us to elucidate the different constituents of the universe such as matter, radiation and dark energy at distinctive cosmic epochs and explore alluring facts about our universe.

One of the striking features of the Freidmann equations is a singularity when $a(t) = 0$. This singularity is the famed Big Bang: the birth of our universe and the beginning of space and time. When $a(t) = 0$ the curvature of spacetime goes to infinity along with the densities of matter and radiation. The Big Bang occurred at every place in space at one single moment in time. The notion of spacetime completely breaks down at the Big Bang singularity along with Einstein's equation making any predictions in cosmology and physics impossible before or at the Big Bang. Therefore, cosmologists' study of the universe begins with the Big Bang, roughly 13.7 billion years ago (Hartle ch. 18, 22).

1.1.2 Dark Matter

When the Universe was in its infancy, cosmic microwave background (CMB) radiation was coupled with baryonic matter. Early universe density perturbations, which

are thought to be the seeds of the large scale structure of the universe, would have been smoothed out by these CMB photons up to scales of $10^{13} \cdot M_{\odot}$. As the Universe expanded this initial small-scale smoothness would have grown to large scales leaving the universe void of any structure. In short, if baryonic matter was the only form of matter in the universe, galaxies, stars, and planets would never have formed (Loeb 30-34).

In 1970, Vera Rubin and Kent Ford observed the orbital speeds of stars in the galaxy M31. The expected orbital velocity of a star around its galactic center is given by the relation

$$v = \sqrt{\frac{GM}{R}} \quad (1.8)$$

where G is the gravitational constant, R is the distance from the galactic center, and M is the mass contained within a sphere of radius R centered at the galactic center. Thus, you expect the relation between orbital speed and orbital radius to be $v \propto 1/\sqrt{R}$. Astonishingly, Rubin and Kent observed motion that was drastically different from Keplerian dynamics; at a large enough radius, orbital velocities of stars stayed roughly constant as radius increased. This was baffling because one would expect these stars to be ejected into interstellar space instead of orbiting the galactic center. This behavior of the non-Keplerian orbital dynamics of stars around galaxies is observed for most galaxies in the Universe. Thus cosmologists deduced that there must be some form of matter surrounding galaxies that provides a sufficient gravitational 'anchor' to avoid these stars from being flung out into intergalactic space. This matter is what is known today as dark matter and the halo surrounding galaxies is the dark matter halo.

There are two theories of dark matter: cold dark matter and hot dark matter. Observations of the large scale structure of the universe indicate that nearly all of the dark

matter in the universe is comprised of cold dark matter. Cold dark matter particles might be supersymmetric particles that were non-relativistic at the time they decoupled from other constituents of the Universe. These particles fall under the category of Weakly Interacting Massive Particles (WIMPs), which are heavy enough to be elusive to modern-day collider experiments and only interact with other particles through gravity and possibly the weak nuclear force. Since dark matter does not interact with electromagnetic radiation, dark matter would be immune to the initial smoothing caused by matter-CMB photon coupling in the early Universe discussed before. As a result, the small-scale density perturbations would not have been smoothed allowing for them to expand and grow into the large scale structure of the Universe we observe today. This makes dark matter a necessary ingredient for cosmological models to succeed in explaining the Universe.

In fact, observations from seven-year data release of the Wilkinson Microwave Anisotropy Probe (WMAP 7) show that the density parameters for dark matter and baryonic matter are $\Omega_{dm} = 0.222 \pm 0.026$ and $\Omega_b = 0.0449 \pm 0.0028$. The most recent observations from the Planck space observatory have observed the density parameters for dark matter and baryonic matter to be $\Omega_{dm} = 0.266 \pm 0.0069$ and $\Omega_b = 0.0449 \pm 0.0007$. The density parameter is defined as

$$\Omega_i \equiv \frac{\rho_i}{\rho_{crit}} \quad (1.9)$$

and ρ_{crit} is the critical density of the universe at the present time given by

$$\rho_{crit} = \frac{3H_0^2}{8\pi G}. \quad (1.10)$$

Thus, most of the matter in the universe is comprised of non-baryonic dark matter. Understanding the nature of dark matter and the ways in which dark matter halos form around galaxies and galaxy clusters is an essential step to understanding the large scale structure

and evolution of the universe (Ryden ch. 8,12).

For a complete model of cosmology one must also consider the vacuum energy, more commonly known as dark energy. Dark energy, denoted Λ , comprises most of the energy density of the universe today and is considered the perpetrator of the accelerated expansion of the universe. This leads us to the widely accepted and tested cosmological model of the universe, that of a Λ CDM cosmology.

1.1.3 Dark Matter Halos

Dark matter halos provide the nurseries in which galaxies and galaxy clusters form, and are the largest constituents of the large scale structure of the universe. Dark matter halos cannot be observed directly, but their existence can be inferred from their gravitational effects on the rotation curves of galaxies and the observed gravitational lensing effects they beget. If dark matter halos in fact did not exist, this would lead one to conclude the Einstein's Theory of General Relativity describing curved spacetime and gravitational interactions is incorrect.

1.1.4 Dark Matter Halo Shapes: Simulations

Through CDM simulations (Springel et al 2005, Klypin et al. 2011) it has been determined that dark matter halos are non-spherical three-dimensional triaxial ellipsoids. Dubinsk & Carlberg (1991) suggest that halos are triaxial ellipsoids and find that the mean halo shape is more spherical at larger radii. In Fig. 1 we show this behavior by plotting a

halo of mass $3 \cdot 10^{13} h^{-1} M_{\odot}$ — we distinctly see a more triaxial inner radius and spherical outer radius. Frenk et al. 1988 find the opposite effect, halos are more spherical at smaller radii and tend to be more triaxial as radius increases. Past work (Warren et al. 1992, Tormen et al. 1997) suggest that the tendency for halos to be triaxial is supported by an internal anisotropic velocity dispersion. Because most halos possess such an inclination to being non-spherical, it leads one to believe that understanding the shapes of dark matter halos will lead to a better understanding of halo formation and galaxy morphology.

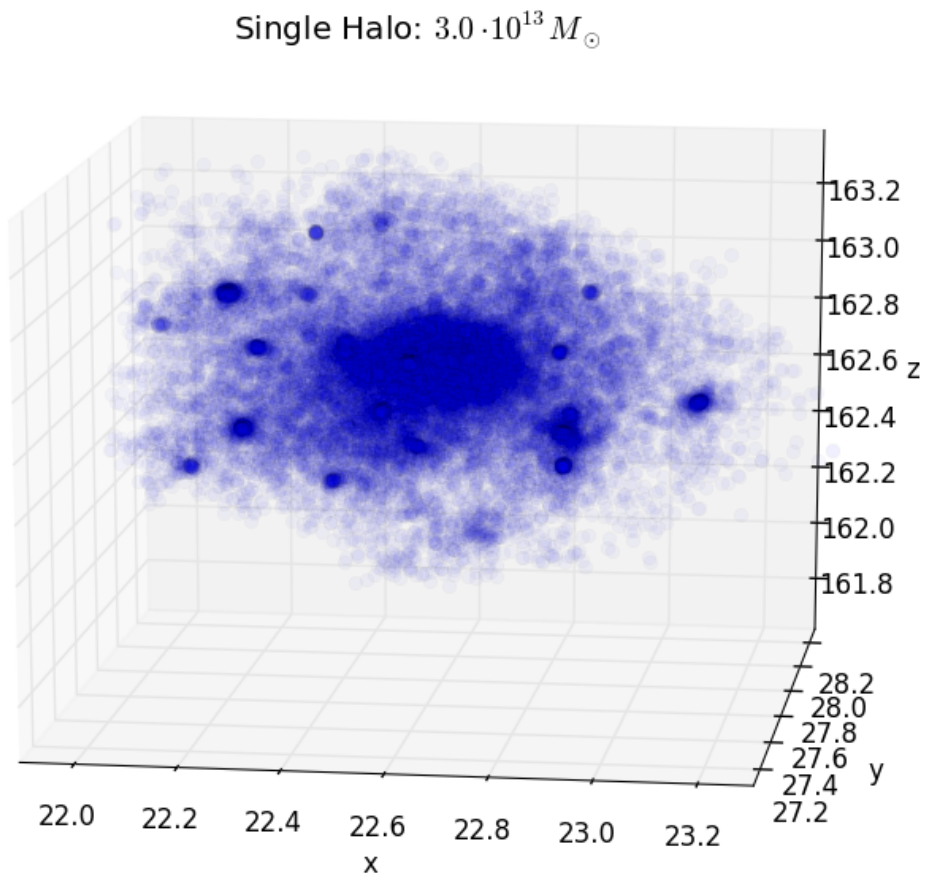


Figure 1.1: Dark matter halo of mass $3 \cdot 10^{13} h^{-1} M_{\odot}$ in the Bolshoi-Planck simulation.

1.1.5 Dark Matter Halo Shapes: Observations

In recent years, improvements in observational techniques have allowed for measurements of the shapes of cluster and galaxy mass halos. Measurements of our own Milky Way halo have found that the halo surrounding our Galactic disk is oblate (Olling et al. 2000) or nearly spherical (Ibata et al. 2001, Majewski et al. 2003). Helmi et al. 2004 and Law et al. 2005 find halos to be prolate by inspecting M giants within the leading stream and finding a best fit prolate halo. Halo shapes measurements at high redshifts can be done using weak lensing studies. Studies by Hoekstra et al. 2004 using data from the Canada-France-Hawaii telescope find halos to be prolate and triaxial.

1.2 In This Thesis

This thesis builds on an immense body of work done studying the structure of cold dark matter halos. With recent advancements in N-body cosmological simulations and supercomputing the methods of studying the structure of dark matter halos has evolved. N-body simulations allow for the opportunity to test Λ CDM with unprecedented accuracy and compare a large spectrum of cosmological parameters with observation.

We use the Bolshoi (Klypin et al. 2011), Bolshoi-Planck and Multidark (Prada et al. 2011) cosmological simulations enabling us to study the concentration and shape of dark matter halos over six orders of magnitude. The Bolshoi, Bolshoi-Planck and Multidark were run on the NASA *Pleides* supercomputer. The halo-finders used in our analysis are the Rockstar Phase-space Temporal Halo Finder (Rockstar; Behroozi et al. 2013) and the Bound Density Maximum (BDM) halo finder (Klypin et al. 1999). Halo catalogues documenting properties of dark matter halos have been produced by both Peter Behroozi

(Rockstar) and Anatoly Klypin (BDM) and are extensively used in this analysis. Another notable halo finder is Friends-of-friends (FOF; Davis et al. 1985), but we restrict this analysis to catalogs computed by Rockstar and BDM.

It is interesting to examine how halo concentration $c \equiv \frac{R_{vir}}{R_s}$ changes as a function of mass and evolves over cosmological time (redshift). We define the concentration by using a scale radius found by fitting a Navarro-Frenk-White density profile and a method presented in Prada et al. 2011. The method used in Klypin et al. 2011 has been shown to be more robust than a fitted scaled radius for halos with 100 particles or less because halo profiles are not well defined for distances on the order of the force resolution of the simulation (Behroozi et al 2013). Recent work has shown that the median concentration steadily decreases with increasing mass and redshift (Bullock et al. 2001). More recently, Klypin et al. 2010 demonstrated that for high redshifts, median concentration starts flat and increases with mass. The most interesting feature of the mass-concentration of halos is we observe the relation to flatten for halo masses $\sim (10^{14} - 10^{15})h^{-1}M_{\odot}$ (Neto et al. 2007, Prada et al. 2011). Explanations for this non-monotonic behavior in higher mass halos have been attributed to the tendency for massive galaxy cluster size halos to be out of dynamical equilibrium, thus having lower concentration (Neto et al 2007 and Ludlow et al. 2012). They claim that if we restrict our analysis to only 'relaxed' halos (halos in dynamical equilibrium) the upturn goes away and the mass-concentration relationship behaves monotonically for the entire mass range. We present our results of the mass-concentration relationship by using the same and more restrictive values of the selection criteria for halo relaxation presented by Neto et al. 2007 and later used by Ludlow et al. 2012. One of the goals of this paper is to further examine the effects of restricting our

analysis to only halos in dynamical equilibrium.

It has been shown halo shapes become more elliptical and non-spheroidal as mass increases (Allgood et. al 2006, Schnieder et al. 2012). Halo shape in N-body simulations can be recovered from the eigenvalues and eigenvectors of a diagonalized shape tensor (Zemp et al. 2011) or diagonalized weighted inertia tensor (Allgood et al. 2006). Past work has shown (Dubinski & Carlberg 1991, Jing & Suto 2002, Allgood et al. 2006) have found that the mean shape of halos is highly correlated to mass. We compare these two common methods for calculating dark matter halo shape.

Section 2 of this thesis discusses the simulations, halo-finders, and halo catalogues used in this work. In section 3 we present results for the mass-concentration relationship of dark matter halos and the effects of halo selection criteria on concentration. In section 4 we present our results for calculating halo shape using two different methods at two different radii. We summarize and discuss the main results of this thesis in section 5.

2

Simulations

Cosmological N-body simulations such as Bolshoi, MultiDark, Bolshoi-Planck and Millennium (Springel et al. 2005) allows us to test the predictions of a Λ CDM cosmology with unprecedented accuracy. More specifically, we can study dark matter halo concentrations and shapes over six orders of magnitude of halo mass to encompass in our analysis galaxy sized halos and galaxy-cluster sized halos. Parameters for these simulations are given in Table 1.

2.1 Numerical Simulations

The Bolshoi, MultiDark and Bolshoi-Planck simulations use cosmological parameters from the five-year WMAP, seven-year WMAP and Planck satellite data release results. See table 1 for these parameters. The Adaptive-Refinement-Tree (ART) code was used for Bolshoi and MultiDark, where ART is an Adaptive-Mesh-Refinement (AMR) algorithm. For more detailed descriptions of the code please refer to Kravstov et al. 1997 and Kravstov 1999. For details of the Bolshoi simulation refer to Klypin et al. 2010.

Table 2.1: Parameters for Λ CDM cosmological simulations. L_{box} is the length of one side of the simulation, N_p is the number of particles in the simulations, ϵ is the force resolution in a comoving (constant in time) coordinate system, M_p is the mass resolution (mass of each particle). The cosmological density parameters for the simulations are: Ω_m , Ω_b , Ω_Λ . The spectral index of the primordial power spectrum is n_s , h is the Hubble constant at the present time in units of 100 km/sMpc^{-1} and σ_8 is the amplitude of mass density fluctuations of spheres of $8 \text{ Mpc}h^{-1}$ at redshift $z=0$. Millennium parameters from Springel et al. 2005, Millennium-II parameters from Boylan-Kochlin et al. 2009, Bolshoi parameters from Klypin et al. 2009 and MultiDark parameters from Prada et al. 2011

Name	L_{box} ($h^{-1}\text{Mpc}$)	N_p	ϵ ($h^{-1}\text{kpc}$)	M_p ($h^{-1}M_\odot$)	Ω_m	Ω_Λ	n_s	h	σ_8
Millennium	500	2160^3	5	$8.61 \cdot 10^8$	0.25	0.75	1.00	0.73	0.90
Millennium-II	100	2160^3	1	$6.89 \cdot 10^6$	0.25	0.75	1.00	0.73	0.90
Bolshoi	250	2048^3	1	$1.35 \cdot 10^8$	0.27	0.73	0.95	0.70	0.82
MultiDark	1000	2048^3	7	$8.63 \cdot 10^9$	0.27	0.73	0.95	0.70	0.82
Bolshoi-Planck	250	2048^3	1	$1.55 \cdot 10^8$	0.30	0.70	0.96	0.70	0.83

The mass of each particle in the Bolshoi, Bolshoi-Planck and MultiDark simulation are: $1.35 \cdot 10^8 h^{-1} M_\odot$, $1.55 \cdot 10^8 h^{-1} M_\odot$ and $8.63 \cdot 10^9 h^{-1} M_\odot$ respectively. Our analysis ranges from halos of mass $\sim (10^{11} - 10^{15}) h^{-1} M_\odot$, so we expect smallest halos to have roughly a thousand bound particles and largest halos to have roughly a million bound particles.

2.2 Halo Finders

The main data sets used for the analysis of large cosmological simulations are halo catalogues. Halo catalogues contain all the relevant intrinsic properties such as position, angular momentum, mass and axial ratios needed for detailed analysis. Dark matter halos for Bolshoi and MultiDark simulations have been identified using the Rockstar Halo Finder (Behroozi et al. 2013) and for Bolshoi and Bolshoi-Planck using the Bound Density Maximum algorithm (Klypin & Holtzman 1997, 1999, Klypin et al. 2011, Riebe et al. 2011).

Rockstar starts by dividing the simulation volume into 3D Friends-of-Friends

(Davis et al. 1985; hereafter FOF) groups for parallelization ease. Then a phase metric is found by dividing particle positions and velocities by the group position and velocity dispersions. A phase-space linking length is then chosen such that 70% of each group's particles are linked together in sub groups. This process repeats for new linking-lengths and substructure. Upon all grades of substructure being identified particles are assigned to the nearest halo in phase space. Once all particles have been assigned to halos unbound particles are removed from halos (Behroozi et al. 2013).

BDM detects maxima in the local density and determines a spherical cut-off for each halo. Subsequently, BDM removes those particles that have characteristic velocities which exceed the escape velocity of the halo. More recent versions have BDM (Riebe et al. 2011) have been able to discern between sub-halos, those which center lie within the virial radius of another halo, and distinct halos, those which center does not lie within the virial radius of another halo. BDM does no analysis for sub-halos. In this paper we will present our results using BDM and Rockstar separately.

3

Dark Matter Halo Concentration

The Navarro-Frenk-White (1995, 1996, 1997; hereafter NFW) density profile describing the density as a function of radius for dark matter halos is given by

$$\rho_{nfw} = \frac{4\rho_s}{\frac{r}{r_s}(1 + \frac{r}{r_s})^2}. \quad (3.1)$$

Here r_s and ρ_s are the scale radius and density respectively, where the scale radius is the radius between an inner r^{-1} density profile and an outer r^{-3} profile (Prada et al. 2011). The NFW profile is a useful tool for fitting halo density profiles of bound particles in simulations. Remarkably, the NFW profile tends to hold for all halos that have been assembled in a hierarchical manner and are approaching virial equilibrium, with little dependence on the specifics of the cosmological model (Neto et al. 2007).

One of the parameters, ρ_s or r_s , in the NFW profile can be replaced by a virial parameter (M_{vir} , R_{vir} , V_{vir}) or a parameter at two hundred times the critical density of the universe (M_{200} , R_{200} , V_{200}) where R is the radius, M is the mass and V is the circular

velocity. The virial radius of a halo can then be defined as the radius of a sphere such that

$$M_{vir} = \frac{4\pi}{3} \Delta_{vir} \rho_{cr}(z) R_{vir}^3 \quad (3.2)$$

and the radius at two hundred times the critical density can be defined

$$M_{200} = \frac{4\pi}{3} \Delta_{200} \rho_{cr}(z) R_{200}^3 \quad (3.3)$$

where $\rho_{cr}(z)$ as the critical density of the universe as a function of redshift and the over-density Δ is

$$\Delta(z) = 18\pi^2 + 82(\Omega_m(z) - 1) - 39(\Omega_m(z) - 1)^2 \quad (3.4)$$

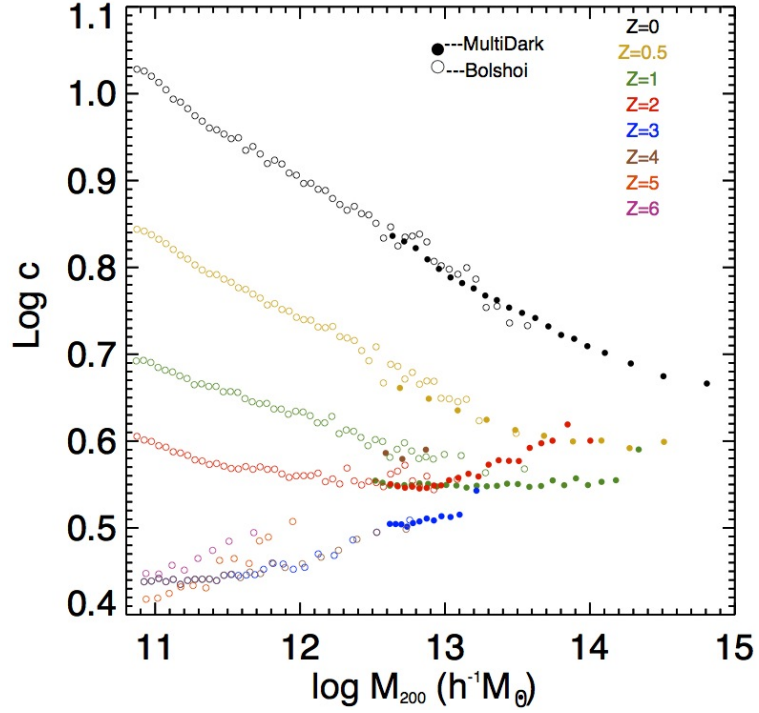


Figure 3.1: Halo mass-concentration relation for distinct halos for Bolshoi (open symbols) and Multidark (filled symbols) the NFW fitted scale radius.

$$c_{vir} = \frac{R_{vir}}{r_s} \quad (3.5)$$

$$c_{200} = \frac{R_{200}}{r_s}. \quad (3.6)$$

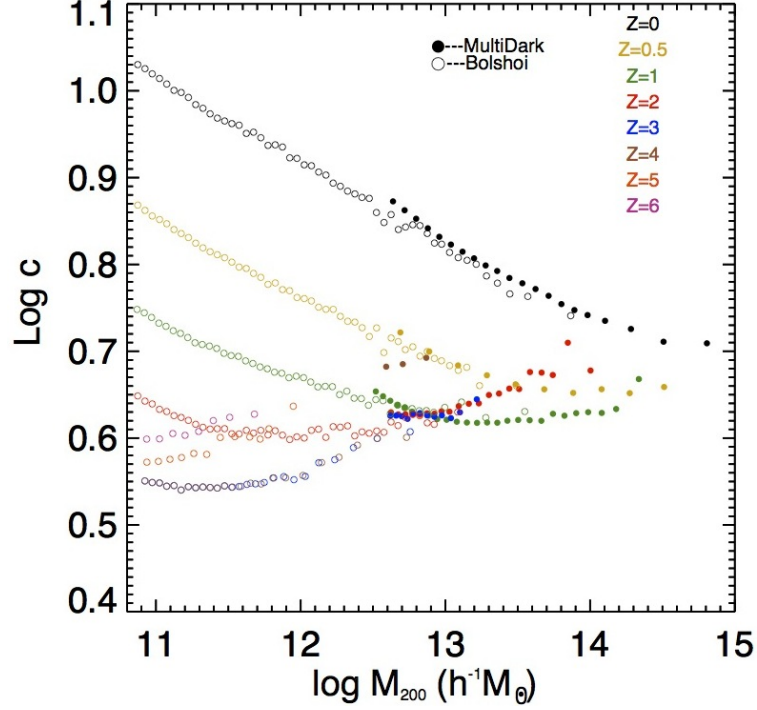


Figure 3.2: Halo mass-concentration relation for distinct halos for Bolshoi (open symbols) and Multidark (filled symbols) using the Klypin scale radius.

3.1 Halo Concentration: Rockstar

Rockstar defines the halo concentration $c = \frac{R_{vir}}{r_s}$ using two different methods to calculate the scale radius. The first method, divides halo particles into a maximum of 50 radial mass bins of equal size with no less than 15 particles per bin (Behroozi et al. 2013) and a NFW profile is used to determine the best fit for the scale radius. Figure 3.1 shows the concentration-mass relationship using this definition of the scale radius.

The second method to calculate the scale radius (Klypin et al. 2011) uses the

maximum circular velocity V_{max} and the circular velocity at the virial radius and notes that these (and any two other independent quantities) can be used to estimate the concentration.

The velocity ratio

$$V_{max} \sqrt{\frac{R_{vir}}{GM_{vir}}} \quad (3.7)$$

can be used to estimate the halo concentration regardless of the halo density profile, but for the case of the NFW profile the velocity ratio is given in Klypin et al. 2011 by

$$V_{max} \sqrt{\frac{R_{vir}}{GM_{vir}}} = \left(\frac{0.216c}{f(c)} \right) \quad (3.8)$$

where $f(c)$ is:

$$f(c) \equiv \ln(1+c) - \frac{c}{1+c}. \quad (3.9)$$

Since we already have the virial ratio for each halo, Eq. 3.8-3.9 are solved numerically to get the concentration $c = \frac{R_{vir}}{r_s}$. We then can easily derive the scale radius, which we call the Klypin scale radius. The concentration used plotting the Klypin scale radius is showing in Fig. 3.2.

For both methods of defining the concentration we see the concentration flatten for higher mass halos and as redshift increases. We see a strong upturn from redshift $z=2$ and higher. The concentration defined by the Klypin scale radius is systematically higher than for the NFW fitted scale radius for all redshifts and mass ranges. Using the NFW fitted scale radius we observe a much steeper upturn for redshifts $z=3-6$.

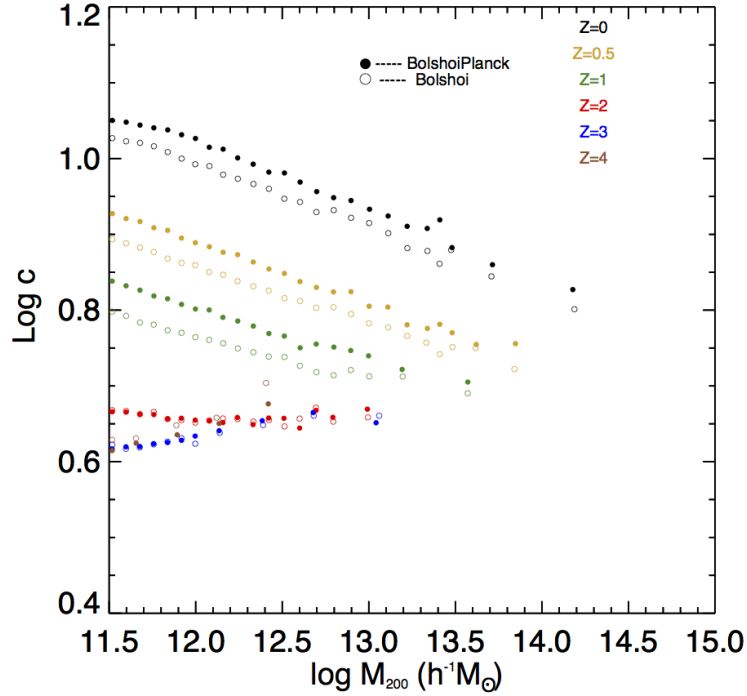


Figure 3.3: Halo mass-concentration relation for distinct halos for Bolshoi (open symbols) and Bolshoi-Planck (filled symbols) using the Klypin scale radius and the BDM halo finder.

3.2 Halo Concentration: BDM

As mentioned before, we restrict our BDM analysis to only the Bolshoi and Bolshoi-Planck simulations leading our analysis for BDM to cover a smaller mass range. We also restrict our analysis to redshifts $z=0-4$. The method which BDM determines the halo concentration is presented in Klypin et al. 2011 where the following equation is solved numerically (see Eq. 3.7-3.9)

$$\left(\frac{0.216c}{\ln(1+c) - \frac{c}{1+c}} \right). \quad (3.10)$$

As expected, the BDM mass-concentration relation (Fig. 3.3) resembles the Rockstar relation (Fig. 3.2) when the Klypin scale radius is used. If we compare the results of BDM and Rockstar for the Bolshoi simulation, we find that the concentration found using Rockstar has a steeper slope for $z=0-2$ and has a more prominent upturn for $z=3-4$.

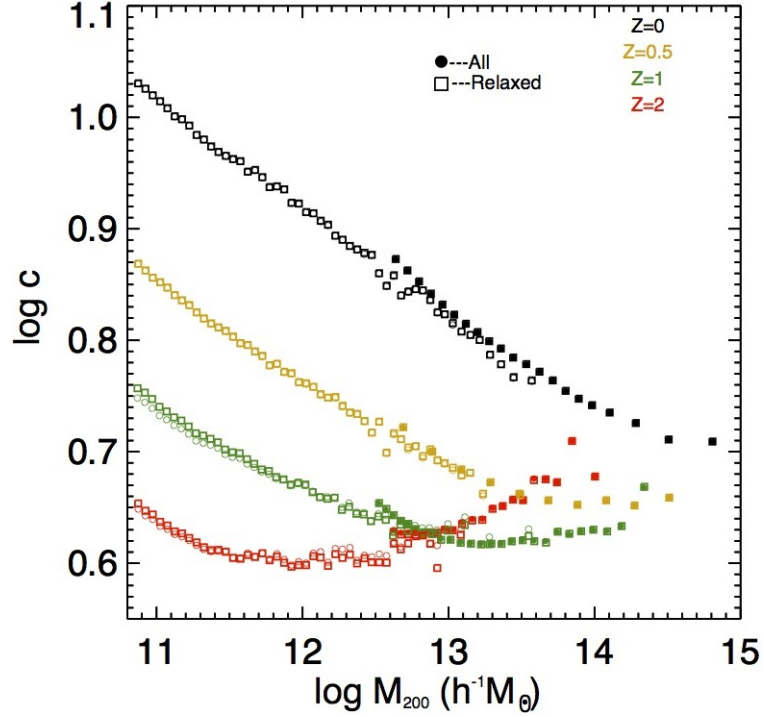


Figure 3.4: Halo mass-concentration comparison between all the halos and the relaxed sample. The relaxation criteria in this figure is $\kappa < 1.3$. Squares represent the relaxed samples and circles represent the whole simulation. Bolshoi is open symbols and Multidark is closed symbols.

3.3 Dark Matter Halo Selection Criteria: Rockstar

Neto et al. 2007 states that because dark matter halos are structures that are constantly evolving due to mass accretion, a large subset of halos tend to be out of virial equilibrium. In addition, because many large halos are undergoing major mergers they tend to have poorly defined centers making spherically defined density profiles not as useful. Recent results (Neto et al. 2007, Ludlow et al. 2012) have found that if we reduce our analysis to only halos in dynamical equilibrium (relaxed halos), the upturn and flattening in concentration shown in Fig 3.1-3 disappears. We use the following two objective criteria (Maccio et al. 2008 and Neto et al. 2007) to define a halo as relaxed:

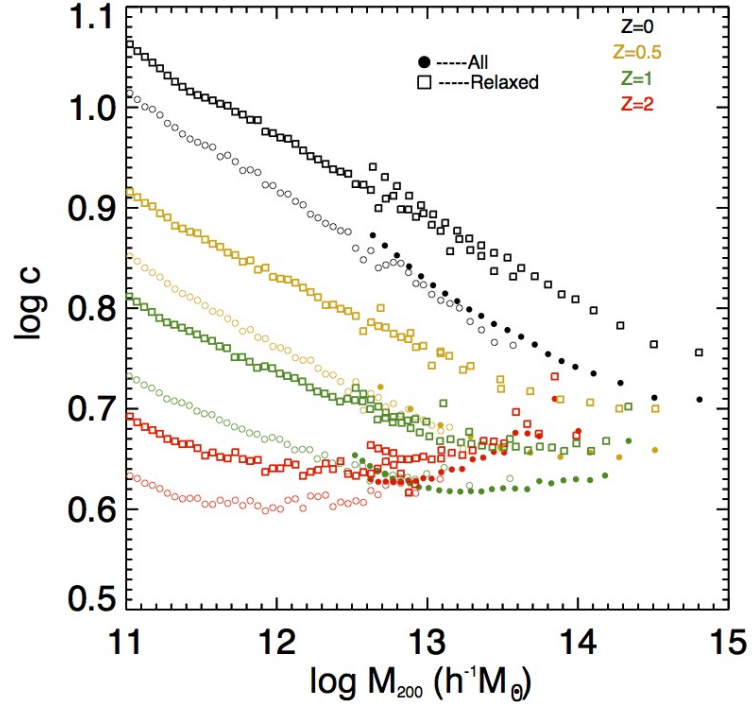


Figure 3.5: Halo mass-concentration comparison between all the halos and the relaxed sample. The relaxation criteria in this figure is $x_{off} < 0.07$. Squares represent the relaxed samples and circles represent the whole simulation. Bolshoi is open symbols and Multidark is closed symbols.

1. Offset Parameter: The offset parameter, x_{off} for each halo is defined as the distance from the particle used as the center of the spherical density profile (potential center), R_p , and the center of mass of the halo, R_{cm} , and normalized by the virial radius

$$x_{off} = \frac{|R_p - R_{cm}|}{R_{vir}}. \quad (3.11)$$

Halos that fall into the relaxation criteria will be radially symmetric and have a low values of x_{off} while halos which are unrelaxed, possibly due to a recent merger, will have asymmetric mass distributions and larger values of x_{off} (Maccio et al. 2008).

We test the values $x_{off} < 0.07$ and $x_{off} < 0.035$ for our 'relaxed' samples.

2. Virial Ratio: The virial ratio (Maccio et al. 2003), which we will denote κ , is a

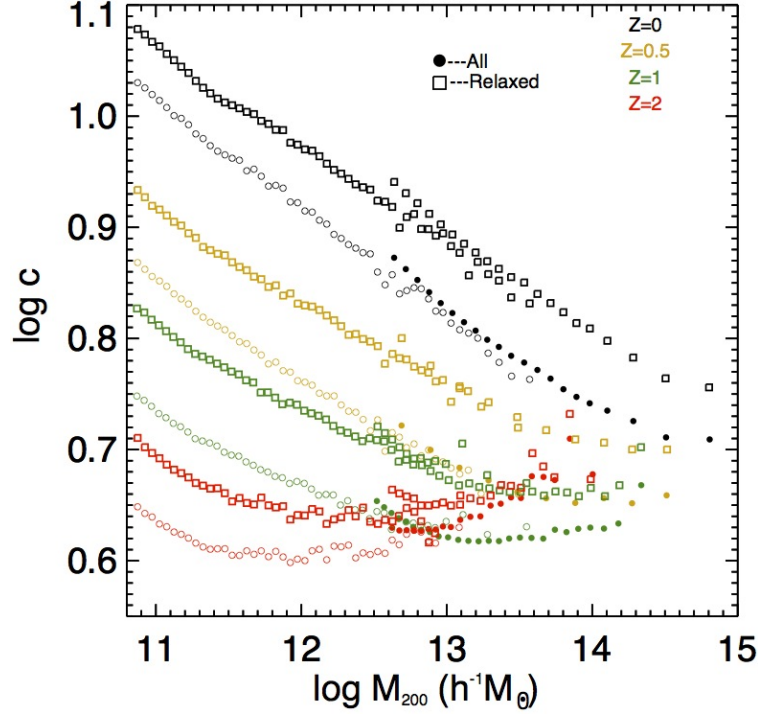


Figure 3.6: Halo mass-concentration comparison between all the halos and the relaxed sample. Both relaxation criteria are used where $\kappa < 1.3$ and $x_{off} < 0.07$. Squares represent the relaxed samples and circles represent the whole simulation. Bolshoi is open symbols and Multidark is closed symbols.

kinematical property of a dark matter halo. We use kinetic energy and potential energies of all the particles within the virial radius to define the virial ratio as follows

$$\kappa = \frac{\sum_{i(r_i < r)} m v_i^2}{\left| - \sum_{i < j (r_{i,j} < r)} \frac{G m^2}{r_{ij}} \right|} = \frac{2T(< R_{vir})}{|-W(< R_{vir})|}. \quad (3.12)$$

We test the values of $\kappa < 1.3$ and $\kappa < 1.1$ in this paper.

We impose the relaxation criteria: $\kappa < 1.3$, $x_{off} < 0.07$ and then both together – shown in Fig. 3.4-6 respectively. These values correspond to those used in Neto et al. 2007 and Ludlow et al. 2011. We see little change in the concentration by restricting the kinematical quantity $\kappa < 1.3$. It is evident that we do not see a tendency towards a more monotonic behavior for halo concentration by restricting our halo samples, but we

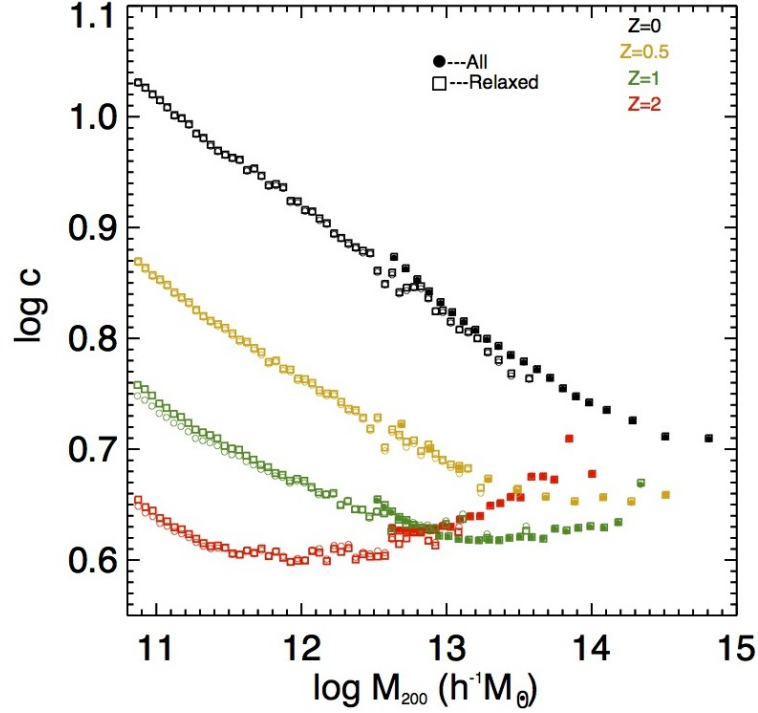


Figure 3.7: Halo mass-concentration comparison between all the halos and the relaxed sample. The relaxation criteria in this figure is $\kappa < 1.1$. Squares represent the relaxed samples and circles represent the whole simulation. Bolshoi is open symbols and Multidark is closed symbols.

do see a systematic increase in concentration for $z=0,0.5,1,2$ when $x_{off} < 0.07$. At redshift $z=2$, we observe the concentrations of the relaxed sample ($x_{off} < 0.07$) converging with the values of our sample containing all halos. By restricting both $x_{off} < 0.07$ and $\kappa < 1.3$ the concentration systematically rises, but there is no evidence of the upturn disappearing. The relaxation criteria x_{off} has a much more dramatic effect on the concentration than κ , which is virtually negligible.

Next we impose more selective relaxation criteria: $\kappa < 1.1$, $x_{off} < 0.035$ and then both together – shown in Fig. 3.7-9 respectively. Restricting our halo sample to halos with $\kappa < 1.1$ (Fig. 3.7) has very little effect on the concentration-mass relation. We do observe a small drop in concentration for redshifts $z=1,2$. If we only sample halos with $x_{off} < 0.035$

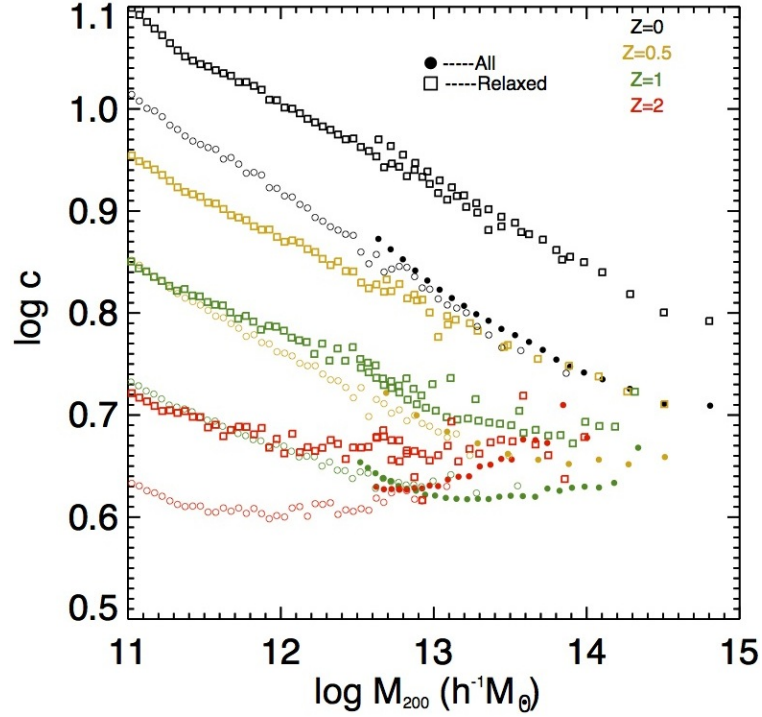


Figure 3.8: Halo mass-concentration comparison between all the halos and the relaxed sample. The relaxation criteria in this figure is $x_{off} < 0.035$. Squares represent the relaxed samples and circles represent the whole simulation. Bolshoi is open symbols and Multidark is closed symbols.

(Fig. 3.8) we see a systematic increase in the concentration for all redshifts. At redshifts $z=1,2$ we observe the concentrations of our relaxed samples ($x_{off} < 0.035$) converge with the values of our sample containing all halos. If we impose both criteria (Fig. 3.9), $x_{off} < 0.035$ and $\kappa < 1.1$, we see the concentration rise significantly more than just restricting $x_{off} < 0.035$ (Fig. 3.8).

3.3.1 Selection Statistics

The percentage of halos which pass our tests for relaxation changes with redshift and mass. In Fig. 3.10 and 3.11 we plot the percentage of halos in each bin which pass a given relaxation criteria. A low percentage (ratio) implies that there are very few relaxed

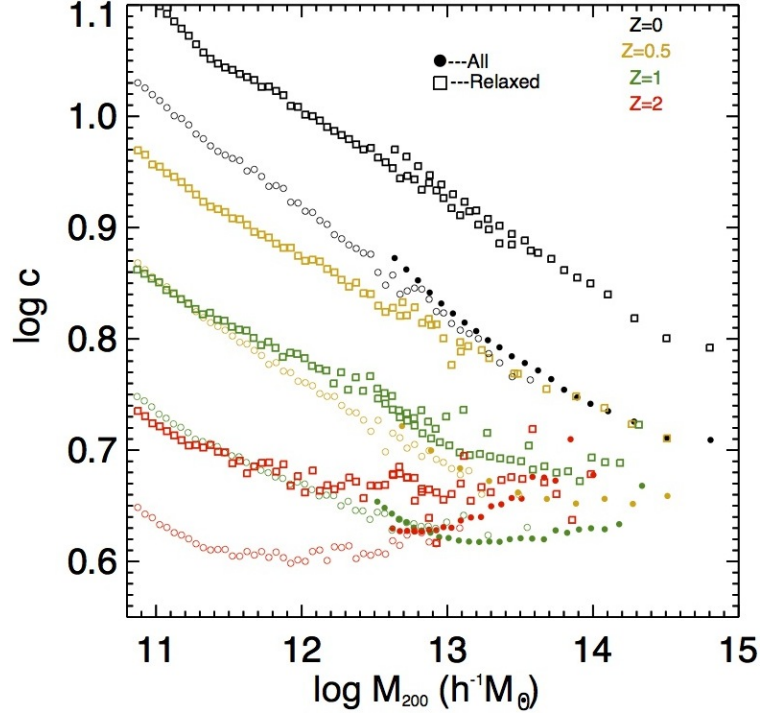


Figure 3.9: Halo mass-concentration comparison between all the halos and the relaxed sample. Both relaxation criteria are used where $\kappa < 1.1$ and $x_{off} < 0.035$. Squares represent the relaxed samples and circles represent the whole simulation. Bolshoi is close symbols and Multidark is closed symbols.

halos in this bin. Massive halos are more likely to be unrelaxed than less massive halos for a given redshift as Fig. 3.10-11 suggest. As redshift increases, the percentage of relaxed halos in a given bin drops significantly. We plot the ratio of relaxed halos to the total number of halos in a mass bin for the relaxation criteria: $x_{off} < 0.07$ (Fig. 3.10) and $x_{off} < 0.035$ (Fig. 3.11). We observe resolution effects for halos with less than 5000 particles: for Bolshoi this is for halos with $M \leq 6.75 \cdot 10^{11} h^{-1} M_{\odot}$ and Multidark $M \leq 4.32 \cdot 10^{11} h^{-1} M_{\odot}$. Vertical lines are drawn for both Bolshoi and Multidark where $N_p = 5000$ in Fig. 3.10 and Fig. 3.11. We do not plot the ratio for our restrictions of κ because nearly all halos in the bins satisfy the criteria. Note that we restrict our samples significantly by imposing these selection restrictions.

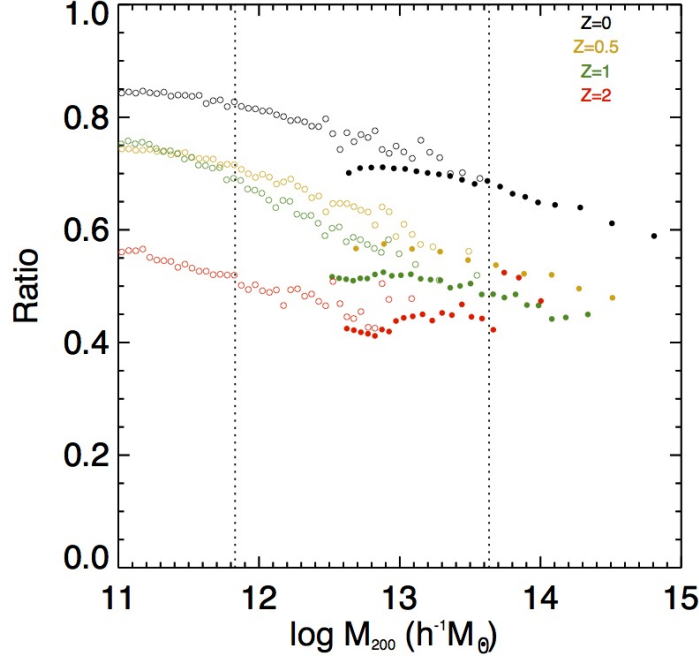


Figure 3.10: Percentage of relaxed halos ($x_{off} < 0.07$) for a given mass bin. Lines are drawn at $N_p = 5000$ for Bolshoi and Multidark. Bolshoi is open symbols and Multidark is closed symbols.

3.4 Dark Matter Halo Selection Criteria: BDM

Our analysis of dark matter halo selection criteria is similar for BDM, but we are analyzing the Bolshoi and Bolshoi-Planck simulations. Our analysis spans the range $z=0-2$. The BDM halos catalogs define the virial ratio slightly differently than Rockstar. The BDM virial ratio κ' is defined as

$$\kappa' = \frac{\sum_{i(r_i < r)} m v_i^2}{\left| - \sum_{i < j (r_{i,j} < r)} \frac{G m^2}{r_{ij}} \right|} - 1 = \frac{2T(< R_{vir})}{|-W(< R_{vir})|} - 1. \quad (3.13)$$

Thus we have the $\kappa' = \kappa - 1$. We analyze the BDM catalogs using the most restrictive selection criteria $\kappa' < 0.1$ and $x_{off} < 0.035$.

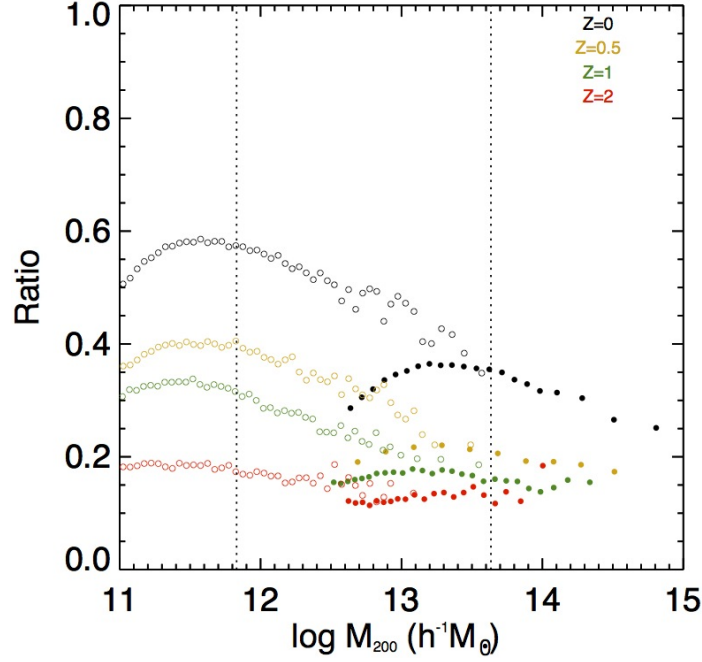


Figure 3.11: Percentage of relaxed halos ($x_{off} < 0.035$) for a given mass bin. Lines are drawn at $N_p = 5000$ for Bolshoi and Multidark. Bolshoi is open symbols and Multidark is closed symbols.

Fig. 3.12-13 show the effects of applying the selection criteria $x_{off} < 0.035$ for both the Bolshoi and Bolshoi-Planck simulations respectively. As seen with the results from the previous section, we see an increase in concentration for the relaxed samples at each redshift. We still observe the upturn at high redshifts, although it is harder to tell because our BDM analysis does not span as large of a mass range as our Rockstar analysis.

Fig. 3.14-15 show the effects of applying the selection criteria $\kappa' < 0.1$ for both the Bolshoi and Bolshoi-Planck simulations. Unlike our Rockstar analysis we see a substantial change in in concentration for our restricted samples — concentration increases substantially at each redshift. We do not see the upturn disappear. Furthermore, this selection criteria reduces our sample to such a small sample size of halos that we begin to see large deviations from the underlying trend in concentration.

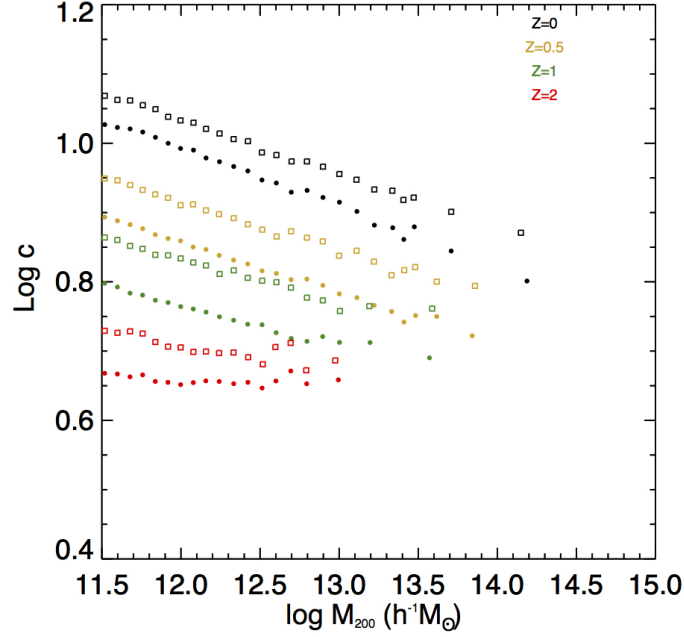


Figure 3.12: Percentage of relaxed halos ($x_{off} < 0.035$) for a given mass bin in the Bolshoi simulation. Squares represent the relaxed samples and circles represent the entire simulation.

3.4.1 Selection Statistics

We now present our selection statistics for the Bolshoi and Bolshoi-Planck simulations using the BDM halo finder. As stated before, the percentage of halos which pass our tests for relaxation changes with redshift and mass. In Fig. 3.16 and 3.17 we plot the percentage of halos in each bin which pass a given relaxation criteria. A low percentage (ratio) implies that there are very few relaxed halos in this bin. Massive halos are more likely to be unrelaxed than less massive halos for a given redshift as Fig. 3.16-3,17 suggest. As redshift increases, the percentage of relaxed halos in a given bin drops significantly. We plot the ratio of relaxed halos to the total number of halos in a mass bin for the relaxation criteria in the Bolshoi and Bolshoi-Planck simulations: $x_{off} < 0.0035$ (Fig. 3.16) and $\kappa' < 0.1$ (Fig. 3.17).

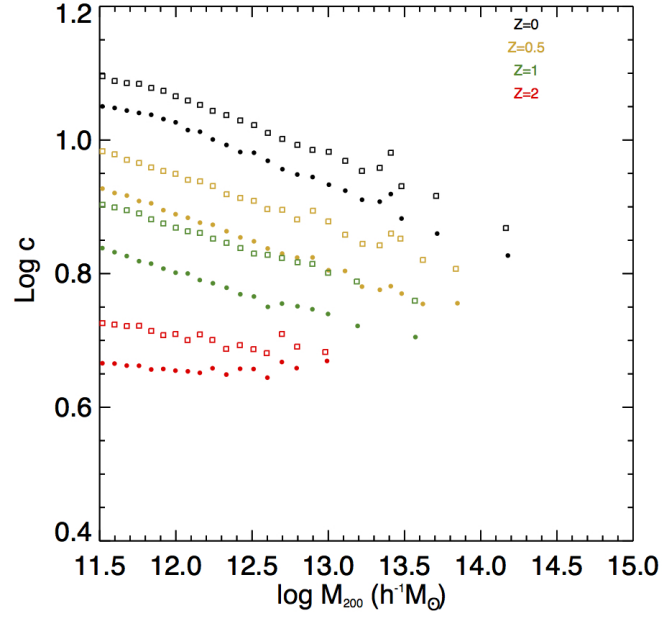


Figure 3.13: Percentage of relaxed halos ($x_{off} < 0.035$) for a given mass bin in the Bolshoi-Planck simulation. Squares represent the relaxed samples and circles represent the entire simulation.

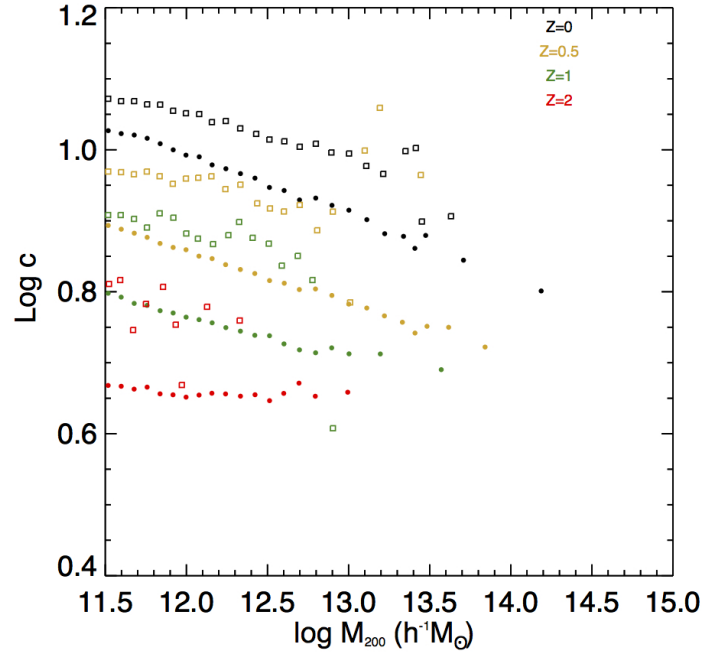


Figure 3.14: Percentage of relaxed halos ($\kappa' < 0.1$) for a given mass bin in the Bolshoi simulation. Squares represent the relaxed samples and circles represent the entire simulation.

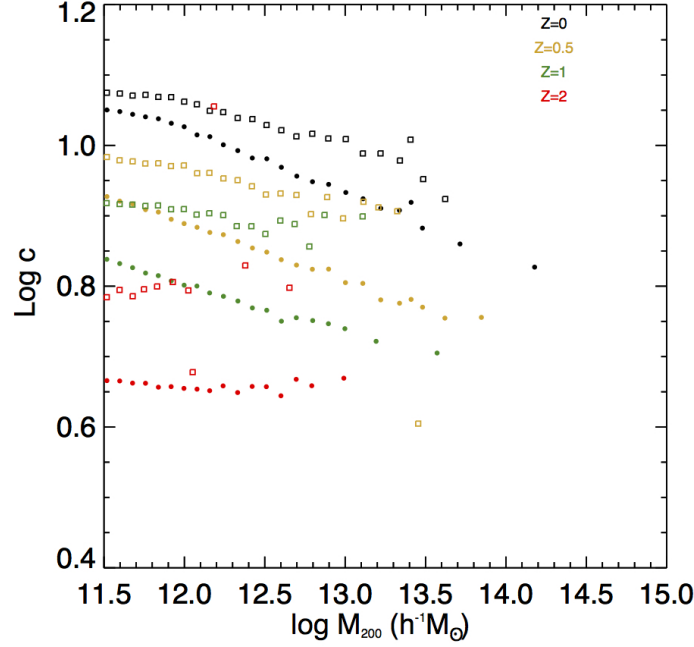


Figure 3.15: Percentage of relaxed halos ($\kappa' < 0.1$) for a given mass bin in the Bolshoi-Planck simulation. Squares represent the relaxed samples and circles represent the entire simulation.

For our restriction $x_{off} < 0.035$ we see the halos that are relaxed are a small subset of the all the halos. The number of relaxed halos in each bin decreases with mass and redshift as expected. We see resolution effects in the low mass range. The Bolshoi simulation has a larger number of relaxed samples at each redshift that satisfy the relaxation criteria $x_{off} < 0.035$. The restriction of $\kappa' < 0.1$ deviates greatly from any other of the selection statistics ratios we have yet seen. The number of relaxed halos in each bin decreases with mass and redshift, but to such a large extent that the number of relaxed halos in each bin become nearly negligible. This is indication that using the selection criteria in this paper is not a promising way to understand the upturn in concentration. The Bolshoi-Planck simulation has a higher number of relaxed samples that satisfy the relaxation criteria $\kappa' < 0.1$.

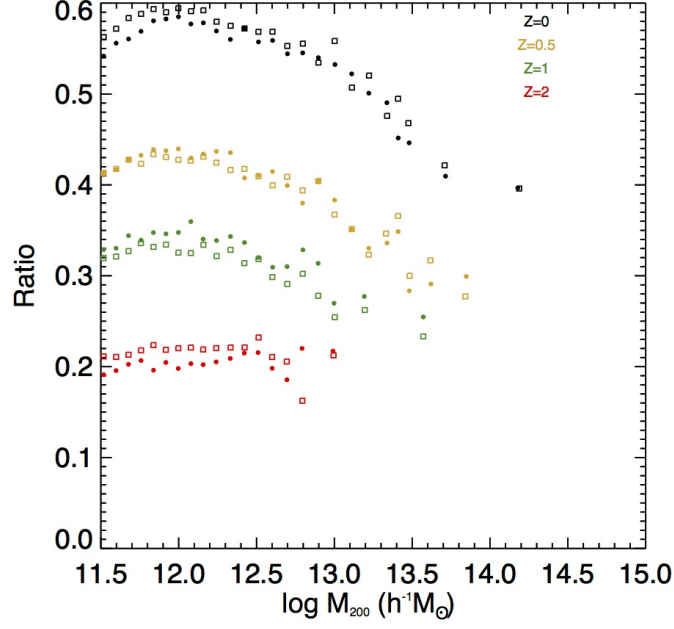


Figure 3.16: Percentage of relaxed halos ($x_{off} < 0.035$) for a given mass bin in the Bolshoi and Bolshoi-Planck simulations. Squares represent the Bolshoi simulation and circles represent the Bolshoi-Planck simulation.

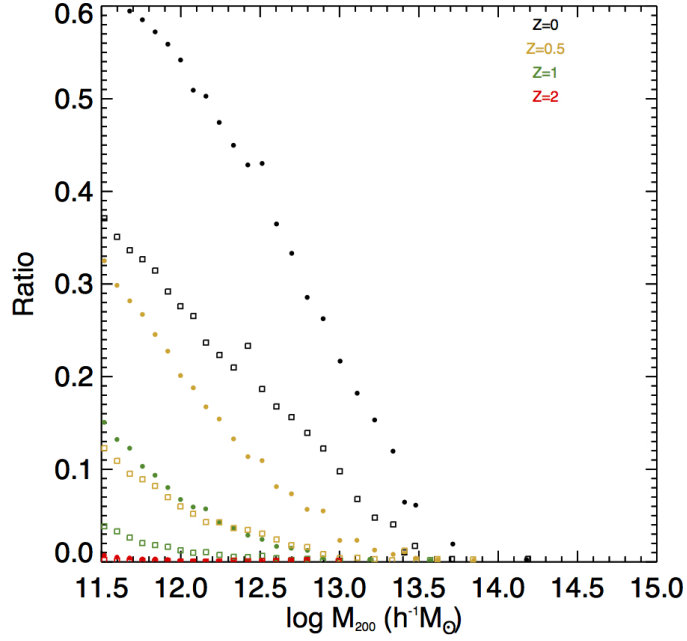


Figure 3.17: Percentage of relaxed halos ($\kappa' < 0.1$) for a given mass bin in the Bolshoi and Bolshoi-Planck simulations. Squares represent the Bolshoi simulation and circles represent the Bolshoi-Planck simulation.

4

Dark Matter Halo Shapes

We investigate the shapes of dark matter halos for the Bolshoi, Bolshoi-Planck and MultiDark simulations. To begin exploring dark matter halo shapes we name the three principal axes of dark matter halos a , b , and c and use the following definitions to determine axial ratios (Allgood et. al 2006)

$$s = \frac{c}{a}, p = \frac{c}{b}, q = \frac{b}{a}. \quad (4.1)$$

Ellipsoids are characterized in three classes corresponding to their axial ratio ranges: prolate ellipsoids with axial ratios $s \approx q < p$, oblate ellipsoids with axial ratios $s \approx p < q$, and triaxial ellipsoids where all three principal axes have different lengths and generally follow $a > b > c$. It has been shown the the median axis ratio $\langle s \rangle$ increases as a function of radius as one moves out farther radii for a given halo and also that clus/ter sized halos tend to be more aspherical than galaxy size halos (Allgood et al. 2006). Peter Behoorizi has calculated shape using a method prescribed by Zemp et al. 2011 and Anatoly Klypin has implemented the Allgood et al. 2006 algorithm with the BDM halo finder. We compare these two methods.

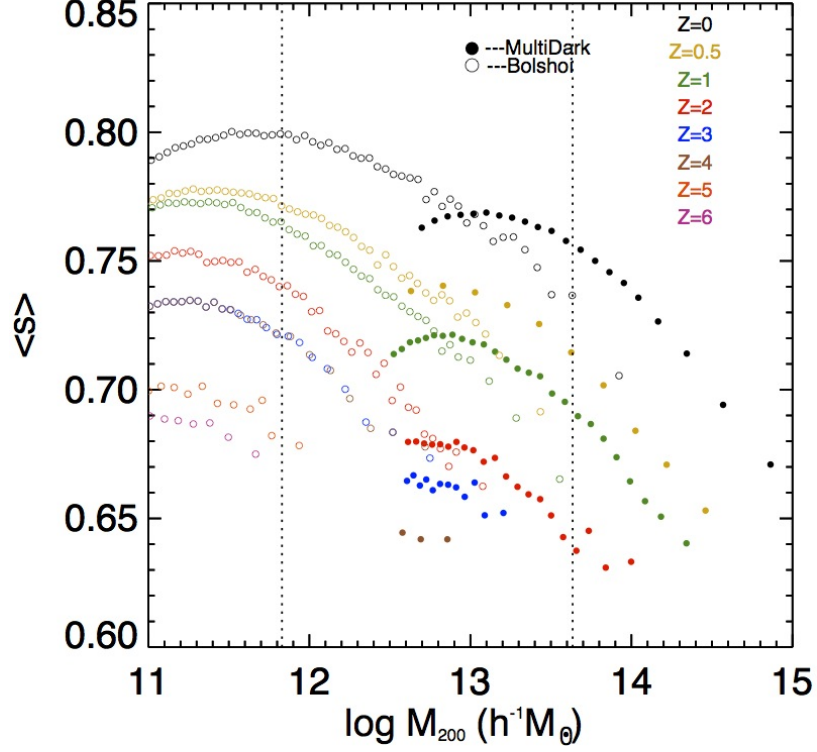


Figure 4.1: Median axis ratio at R_{vir} using the Zemp et al. 2011 method. We indicate where $N_p = 5000$ for Bolshoi and Multidark. Bolshoi is open symbols and Multidark is closed symbols.

4.1 Rockstar Shape Calculation - Zemp et al. 2011

Peter Behroozi has calculated halo shapes using a form of the Zemp et al. 2011 prescription in his Rockstar halo catalogs. As mentioned before, immense motivation for this analysis is held in the method which halo shapes are determined.

$$S_{ij} = \frac{\sum_n m_n x_{i,n} x_{j,n}}{\sum_n m_n} \quad (4.2)$$

Here m_n is the mass and $x_{i,n}$ is i -th component of the position vector of the n -th particle with the summation being done over all particles within the integration volume. A

weight function can be added such that

$$S_{ij} = \sum_n \frac{w(\mathbf{x}) m_n x_{i,n} x_{j,n}}{m_n} \quad (4.3)$$

where $w(\mathbf{x})$ is the weight function dependent on the position vector $\mathbf{x}$ with respect to the center of the halo. It must be noted that if $w(\mathbf{x}) = r_n^{-2}$, we recover equation 4.4. The Rockstar catalogues use $w(\mathbf{x})=1$. The weight function and its effect on the shapes of halos is discussed extensively in Zemp et al. 2011.

The calculation of shape starts by using a spherically symmetric volume. The shape tensor S is diagonalized and we recover the eigenvectors and eigenvalues. As before, the eigenvectors give the directions of the principal axes and the ratio of the eigenvalues allows us to determine the axial ratios s, p, q . The longest principal axis is kept equal to the initial spherical virial radius for all iterations, but is able to rotate freely in space. Then S is calculated again summing over all particles within the new deformed integration volume. This iterative process continues until convergence is reached and we have recovered the triaxial ellipsoid shape. All substructure is excluded in the calculation (Zemp et al 2011).

We see our results for the median axis Ratio using the Zemp et al. 2011 method in Fig. 4.1. We observe resolution effects in the low mass range. The median axis ratio decreases with increasing redshift and mass which agrees with Zemp et al. 2011 and Allgood et al. 2006.

4.2 BDM Shape Calculation - Allgood et al. 2006

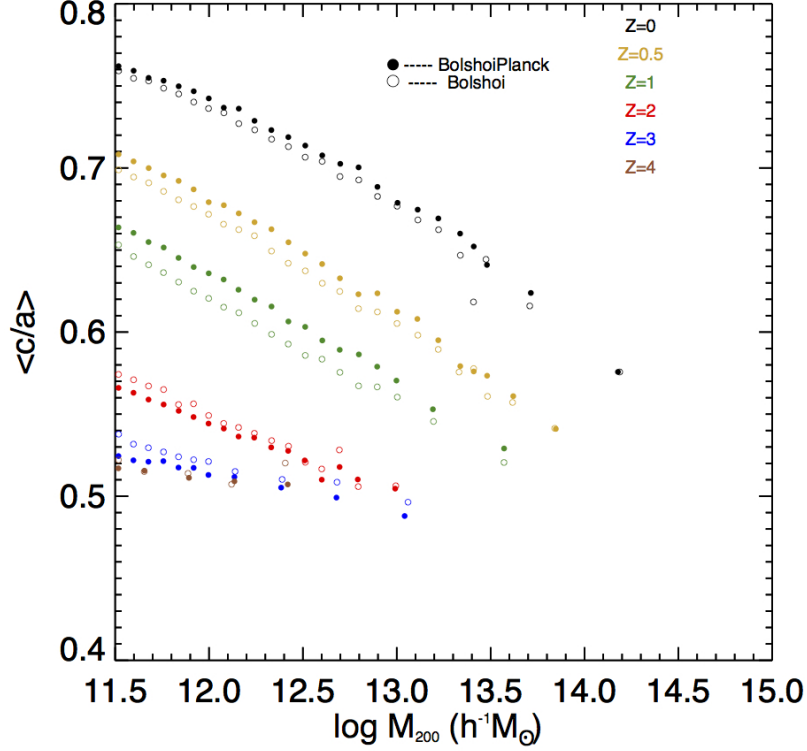


Figure 4.2: Median axis ratio at R_{vir} using the Allgood et al. 2006 method. Bolshoi and Multidark. Bolshoi is open symbols and BolshoiPlanck is closed symbols.

The ellipticity of halos can be found by using eigenvectors and eigenvalues from a diagonalized inertia tensor. The eigenvectors will correspond to the direction of the principal axes in space and the eigenvalues will correspond to the axial ratios s , p , q defined previously. The inertia tensor used is a weighted inertia tensor

$$\tilde{I}_{ij} = \sum_n \frac{x_{i,n} x_{j,n}}{r_n^2} \quad (4.4)$$

where

$$r_n = \sqrt{x_n^2 + \frac{y_n^2}{q^2} + \frac{z_n^2}{s^2}} \quad (4.5)$$

is the elliptical distance to the n -th particle in an eigenvector coordinate system. The eigen-

values $(\lambda_a, \lambda_b, \lambda_c)$ allow us to determine the axial ratios s, q, p where the principal axes are

$$(a, b, c) = \sqrt{\lambda_a, \lambda_b, \lambda_c}.$$

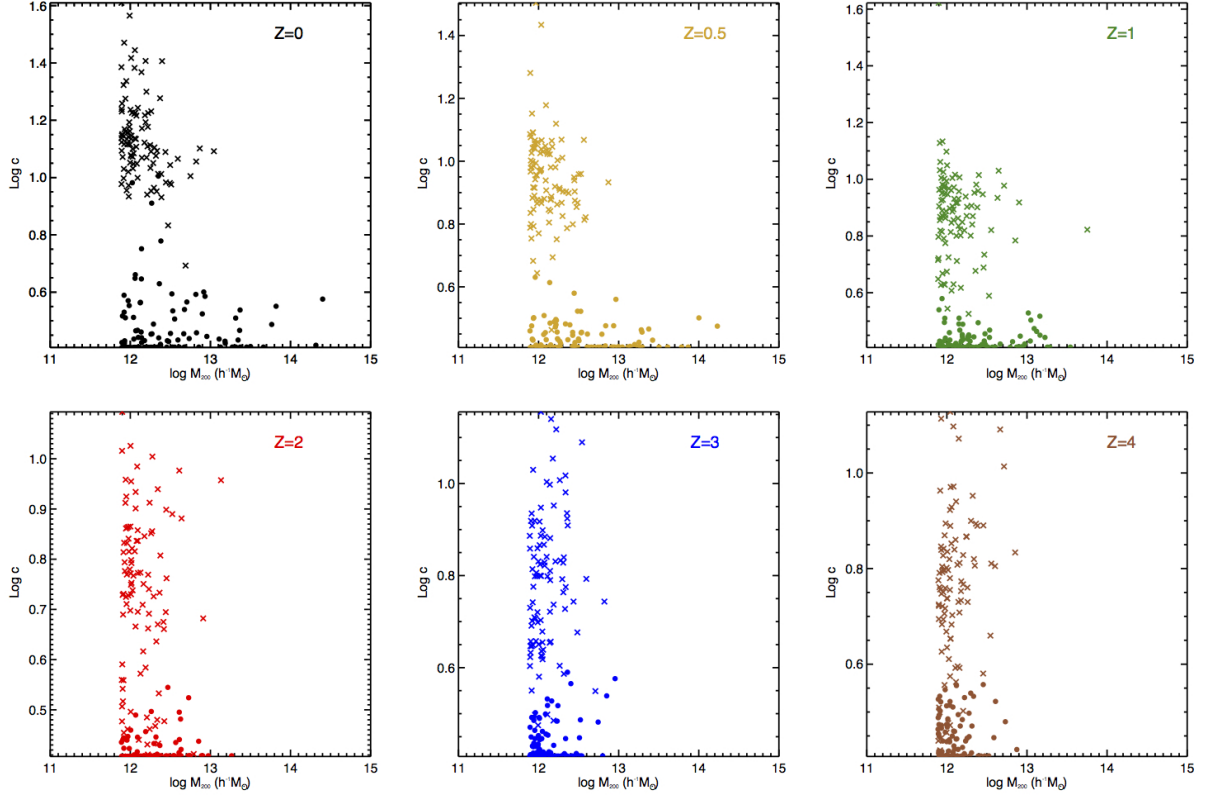


Figure 4.3: Concentration-Mass relation for the 100 most elongated (circles) and 100 least elongated (crosses) halos at each redshift in the Bolshoi-Planck simulation.

This method starts by determining the inertia tensor with $s = q = 1$, a sphere, and including all particles within some determined radius. The initial inertia tensor is deformed along the eigenvectors while keeping the the longest eigenvector constant at the original spherical radius through iterations. The eigenvalues of the inertia tensor are rescaled by:

$$\frac{r_{sph}}{(abc)^{\frac{1}{3}}} \quad (4.6)$$

where r_{sph} is the initial spherical radius. After each iteration, if particles are no longer

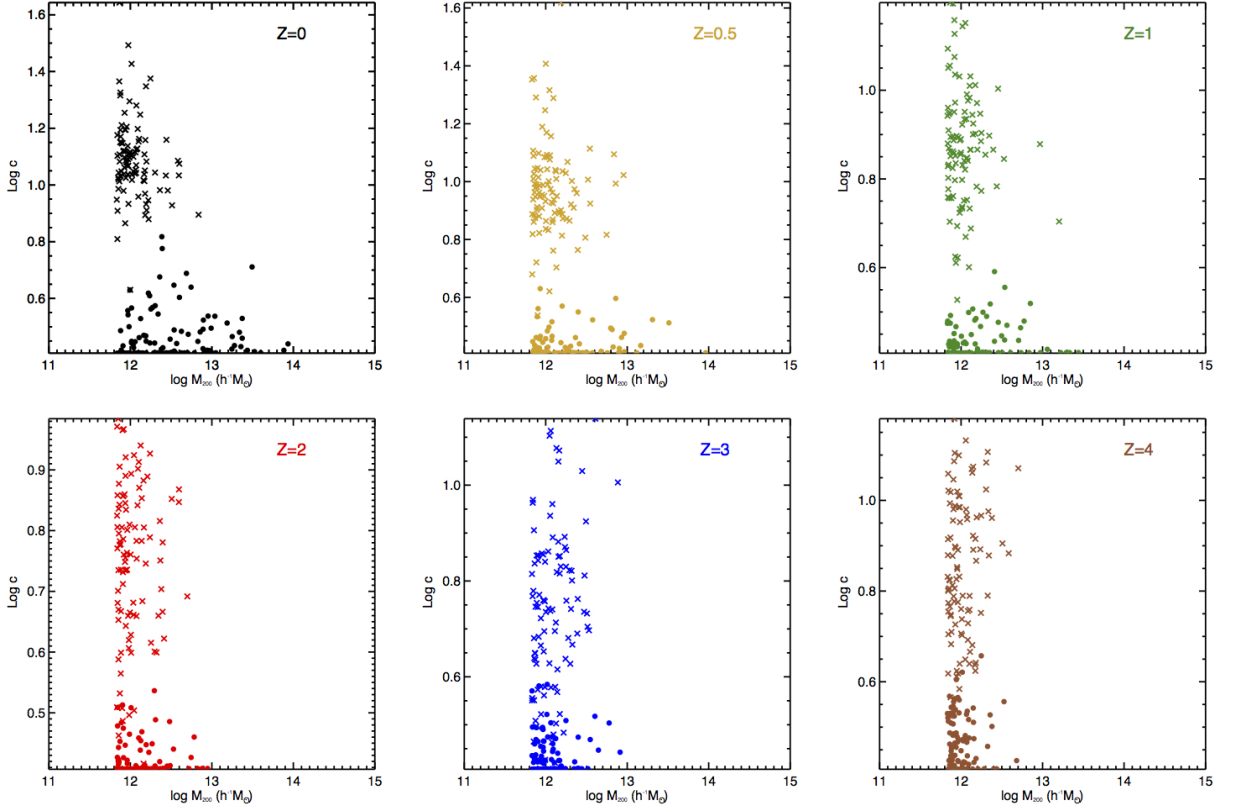


Figure 4.4: Concentration-Mass relation for the 100 most elongated (circles) and 100 least elongated (crosses) halos at each redshift in the Bolshoi simulation.

within the boundary of the halo they are discarded and particles now within the boundary of the halo are included in the subsequent iteration. We repeat this process until the variance of s and q between iterations is less than some predetermined tolerance (Allgood et al. 2006, Schneider et al 2012). Motivation for weighting the inertia tensor (eq. 4.4) by the elliptical radius (eq. 4.5) is to give particles at larger radii less weight. Substructure is not removed.

We plot the median axis ratio versus the halo mass using the Allgood et al. 2006 prescription in Fig. 4.2. We observe the axial ratios monotonically decrease with increasing redshift. The median axis ratio for Bolshoi-Planck simulation is higher for redshifts $z=0-2$. The median axis ratio for the Bolshoi simulation is higher for redshifts $z=3-4$. We don't not

observe the resolution effects seen using the Zemp et al. 2011 method suggesting the Allgood et al. 2006 method is more robust. This becomes increasingly important if one would like to determine the median axis ratio at smaller radii. A probable reason Zemp et al. 2011 experiences resolution effects in the low mass ranges over all redshifts is because it excludes substructure. Excluding substructure will decrease the number of particles within a halo making it harder for the algorithm to resolve a shape. Furthermore, excluding substructure does not allow for comparison with observational dark matter halo measurements.

In Fig. 4.3-4 we plot the concentration of the 100 most spherical and elongated halos in the Bolshoi-Planck and Bolshoi simulations respectively. We find that the most spherical halos are also those with the highest concentration and the most elongated halos are those with the lowest concentration. We observe this trend over all redshifts and for both the Bolshoi and Bolshoi-Planck simulations. This is somewhat surprising because given our spherical definition of the scale radius of a halo, more elongated halos should have higher concentrations because the scale radius would be systematically underestimated leading to higher concentrations. This more importantly suggests that the upturn in concentration seen at all redshifts is infact not a kinematical property of dark matter halos, but has to do with the shape of the dark matter halos

For more detailed discussion on determining shapes of dark matter halos please refer to Dubinski & Carlberg 1991, Jing & Suto 2002, Allgood et al. 2006, Zemp et al. 2011 and Schneider et al. 2012. For discussion of the iteration method refer to Katz 1991, Dubinski & Carlberg 1991, and Warren et al. 1992.

5

Summary and Discussion

In this paper, we examined the shapes and concentrations of dark matter halos in a Λ CDM cosmology. Below we summarize and discuss our major results:

- i. The Rockstar Phase-Space Temporal halo finder and the Bound Density Maximum halo finder were compared by analyzing the concentration and shapes of halos. We found the mass-concentration relationship for halos in the Bolshoi, Bolshoi-Planck and Multidark simulations to agree with the findings of Prada et al. 2011 for both halo finders.
- ii. We applied different selection criteria initially prescribed by Neto et al. 2007 to our halo samples. This had little effect on the upturn in halo concentration, we found it only increased the average concentration over all mass ranges and redshifts. This disagrees with the finding of Neto et al. 2007 and Ludlow et al. 2011.
- iii. The halo selection criteria proposed by Neto et al. 2007 was found to reduced our sample space by such a large amount that we encountered very poor statistics and

large fluctuation from the underlying trend in concentration. Furthermore, most dark matter halos, including the one surrounding our own Milky Way galaxy, are not relaxed. Thus understanding the behavior of halos which are deemed relaxed, we do not gain insight into the properties of the majority of dark matter halos.

- iv. We find the shapes of dark matter halos using the Zemp et al. 2011 method for the Bolshoi and Multidark simulations. The median axis ratio of dark matter halos decrease with increasing halo mass and redshift. We observe resolution effects in the low mass ranges due to small numbers of particles. We attribute this to the fact that substructure is not included in the shape calculation. Excluding substructure will provide poor results if shapes are to be calculated at smaller radii than $r = R_{vir}$, which would not allow for comparison with observational dark matter halo measurements.
- v. The Allgood et al. 2006 method is used to determine the shapes of halos in the Bolshoi and Bolshoi-Planck simulations. We find the median axis ratio of dark matter halos to decrease with increasing halo mass and redshift. Substructure is included in the Allgood et al. 2006 method and we do not observe the resolution effects found using the Zemp et al. 2011 method. We propose to use the Allgood et al. 2006 method in future studies, by including substructure it will perform better at smaller radii making it much more comparable to observations.
- vi. We find that the most elongated halos in the Bolshoi and Bolshoi-Planck simulations are also those with the lowest concentrations and the most spherical halos are those with the highest concentrations. We thus propose that the upturn in halos is not due to a kinematical property of halos, but is in fact caused by the shapes of dark matter

halos.

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